



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CST) (CBCS) Degree Examinations
 January 2023

22CST101 - MECHANICS OF DEFORMABLE BODIES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. Derive equilibrium equations in polar coordinates. | 20 | L*3 | 1 | 3 |
| 2. The components of stress at a point are given by:
$\sigma_x = x^2y + 20$ $\tau_{xy} = 3x^2y$ $\tau_{zx} = xy$
$\sigma_y = x^2z + y^2$ $\tau_{yz} = yz$
$\sigma_z = yz^2 + 10$
Determine whether these components of stress satisfy the equilibrium equations or not at the point (1, 2, 3). If not then determine the suitable body force required at this point so that these stress components become under equilibrium. | | | | |
| | 20 | L3 | 1 | 3 |

Unit – II

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|--|----|----|---|---|
| 3. The stress components at a point are: $\sigma_x = 2$ Mpa, $\sigma_y = 1.5$ Mpa, $\sigma_z = 0$, $\tau_{xy} = 1$ Mpa, $\tau_{yz} = 1$ Mpa, and $\tau_{zx} = -1$ Mpa. Determine the normal and shearing stresses on the plane whose direction cosines are $1/\sqrt{3}$, $1/\sqrt{3}$, $1/\sqrt{3}$. Also calculate the direction of the shear stresses. | 20 | L3 | 2 | 3 |
| 4. The state of stress at a point is given by the following array of terms:
$\begin{matrix} 10 & 5 & 8 \\ 5 & -2 & 4 \\ 8 & 4 & 6 \end{matrix}$ Determine the principal stresses and directions of σ_1 . | 20 | L3 | 2 | 3 |

Unit – III

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|---|----|----|---|---|
| 5. Given the stress function $\phi = - (F/h^3) xy^2(3h-2y)$ where F= constant. Determine the stress components and sketch their variation included in $y=0$, $y=h$, $x=0$ on the side x '+ve' | 20 | L3 | 3 | 3 |
| 6. Derive the stress components for pure bending of a beam and hence obtain an expression for moment and its location from top fibre. | 20 | L3 | 3 | 3 |

Unit – IV

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|---|----|----|---|---|
| 7. Derive the Torsion equation in terms of warping strain ψ and stress function ϕ | 20 | L3 | 4 | 3 |
| 8. Derive an expression for deformation of a prismatic bar, when it is stretched under it's own weight. | 20 | L3 | 4 | 3 |

Unit – V

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|--|----|----|---|---|
| 9. List the theories of elastic failure. Explain any three. | 20 | L3 | 5 | 3 |
| 10. Draw the stress strain curve for different types of materials and explain. | 20 | L3 | 5 | 4 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome